

On the Mechanical Origin of Quantum Statistics

Emergence of Born's Rule and Pauli-Type Exclusion

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Source code: <https://github.com/Zobeewan/emergent-quantum-statistics.git>

Abstract

This document presents numerical simulation results of a local hidden-variable model inspired by hydrodynamic quantum analogs of the “walking droplet” type. Unlike standard pilot-wave approaches in which the guiding field influences the particle without being affected in return, we implement here a continuous bidirectional coupling: the particle acts as a moving source that generates its own pilot field. We show that this local deterministic dynamics with stochastic noise is sufficient to produce, without additional axioms, the emergence of at least two key statistical features of quantum mechanics: relaxation toward the Born distribution ($\rho(x) \approx |\psi|^2$), and a Pauli-type exclusion for two particles interacting through the medium.

1. Physical Model

The system is modeled as the coupling between a complex scalar field $\psi(x, t)$ and N point-like particles with trajectories $x_p(t)$

1.1. Field Dynamics (The Medium)

The pilot wave evolves according to a nonlinear diffusive Schrödinger-type equation with source terms, inspired by the physics of walking droplets:

$$\frac{\partial \psi}{\partial t} = (D_\psi + i\omega)\nabla^2 \psi - \gamma\psi - iV(x)\psi + \sum_k S(x - x_k(t))$$

- D_ψ, ω : Diffusion and dispersion coefficients (analogous to $\frac{\hbar}{2m}$).
- γ : Dissipation (possibly zero for a system with infinite memory and range, but useful for numerical stability).
- $V(x)$: External potential

- **S(x)**: Gaussian source term centered on each particle (particle-to-field feedback)

$$S(x - x_k) = A \exp \left(- \frac{|x - x_k|^2}{2\sigma^2} \right)$$

With:

- **A** : Source emission amplitude
- **σ** : Spatial width of the source

1.2. Particle Dynamics (Guidance)

Particles follow an overdamped Langevin dynamics guided by the phase gradient of the total wave field:

$$dx_p = [\alpha \cdot W(|\psi|^2) \cdot \nabla \phi] dt + \sqrt{2 D_x} dt$$

- **∇φ**: local phase velocity.
- **α**: Inertial coupling coefficient (analogous to $\frac{k}{m}$).
- **W**: Nonlinear saturation factor (prevents singularities at nodes).

$$W(|\psi|^2) = \frac{|\psi|^2}{|\psi|^2 + \epsilon^2} \quad \text{with } (\epsilon = 10^{-4})$$

- **D_x**: Stochastic noise intensity.

$$D_\psi = 0.9 ; \omega = 2.0 ; \gamma = 0.045 ; A = 0.57 ; \alpha = 4.0 ; D_x = 0.28$$

Values currently used, but all model parameters can still be optimized via algorithmic exploration (but computation time is very long).

*The stochastic noise **D_x** is used to simulate chaotic behavior that should emerge from a model with more degrees of freedom and more realistic interactions.*

All simulations are performed in sufficiently large domains so that boundary effects can be neglected.

2. Results: Emergence of Born's Rule (1D)

A single particle is simulated many times over 40,000 time steps. The empirical probability density $\rho(x)$, obtained from the histogram of particle positions, is compared to the time-averaged field intensity $|\psi_{eff}|^2$

Observations:

- The particle, initially launched at random, relaxes toward an ergodic trajectory that samples the eigenmodes of the dynamical cavity it generates.
- The measured correlation between $\rho(x)$, and $|\psi|^2$ is excellent, with a very small relative L^1 error.
- These results support the “quantum relaxation” hypothesis: $|\psi|^2$ acts as a natural statistical attractor of the chaotic dynamics.

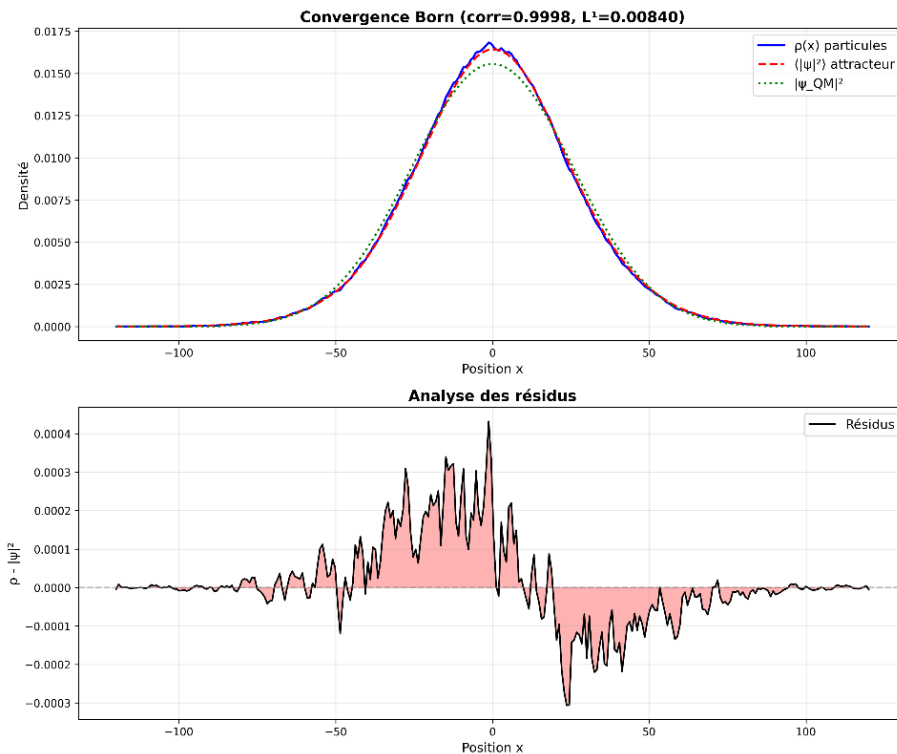


Figure 1. Empirical probability density $\rho(x)$ (blue) versus pilot-wave intensity $|\psi|^2$ (red).

The convergence is nearly perfect (Corr = 0.9998), with a very small error ($L^1 = 0.0084$).

+ Comparison with the analytical solution of the Schrödinger equation (green) is shown for validation.

$N=10,000$ simulated particles.
(Convergence appears rapidly ($N > 600$), and increases monotonically with bigger sample size)

3. Results: Pauli-Type Exclusion (Two-Body Interaction)

The model is extended to two identical particles. In each run, two particle pairs are generated:

- One real pair, capable of interacting indirectly via the deformation of a shared field, coupling: $\psi = \psi_1 + \psi_2$
- One “ghost” pair evolving in separate fields and therefore unable to interact.

Spatial correlations are then analyzed to detect fermion-like signatures.

No antisymmetrization of the wave function is imposed. The observed exclusion arises solely from dynamically induced field-mediated interactions.

Observations:

- **Spatial correlations – Fermi hole:** The pair correlation function $g^{(2)}(r)$ drops below unity at short distances, with $g^{(2)}(0) < 0.35$, indicating strong exclusion behavior.
- **Spatial exclusion:** Particles spontaneously avoid each other. The joint position heatmap shows a suppression along the diagonal $x_1 = x_2$. The density of close encounters is significantly reduced for interacting pairs compared to non-interacting ghost pairs.

Page 2: Spatial Correlations and Pauli Exclusion

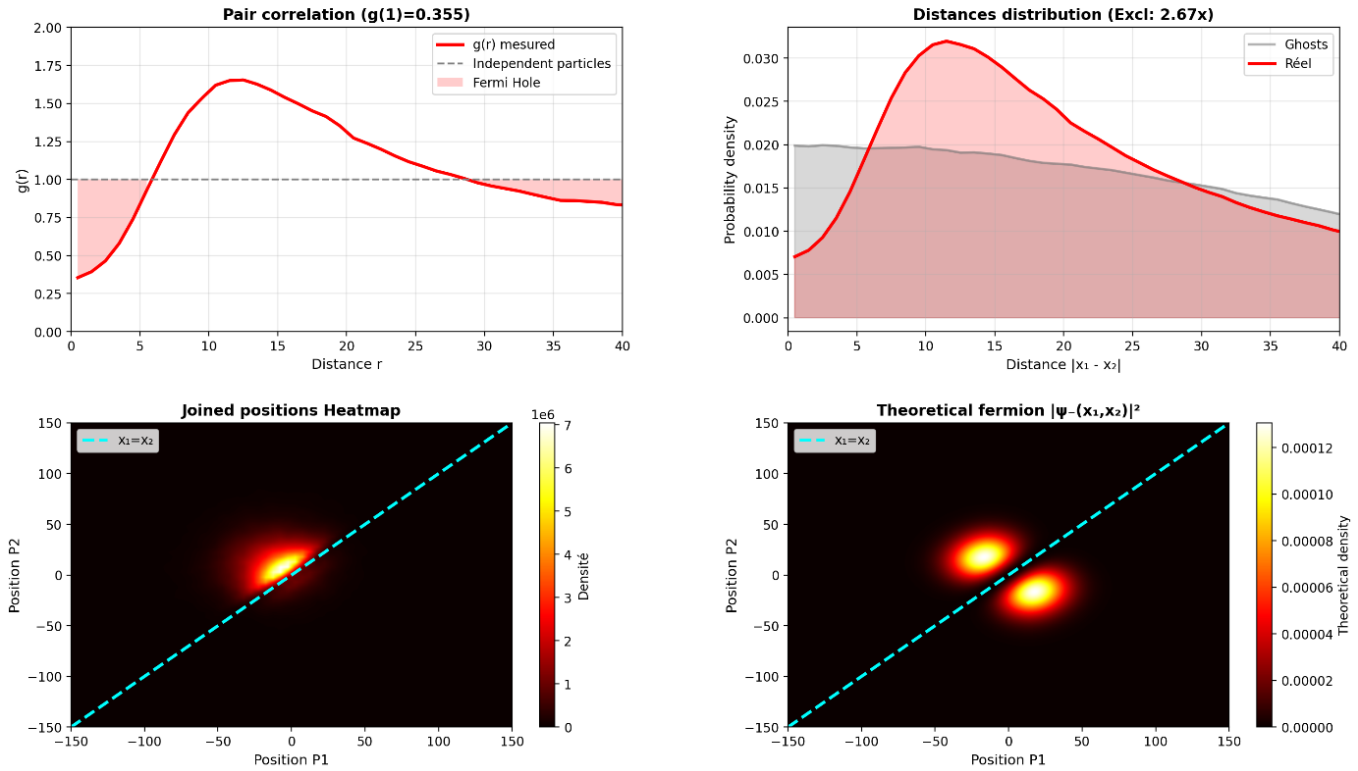


Figure 2. Simulations over 12,000 particle pairs.

Two-particle correlations: (top-left) $g^{(2)}(r)$ showing a Fermi-hole-like structure; (bottom-left) joint position heatmap (x_1, x_2) with a depleted diagonal, illustrating Pauli-type exclusion.

Interpretation of Quantitative Differences:

The asymmetry observed in the joint position heatmap is not a failure of prediction or a failure of exclusion, but a direct consequence of using distinguishable particles with fixed initial labels.

The discrepancy between the measured heatmap (bottom-left) and the theoretical fermionic heatmap (bottom-right) arises from particle distinguishability in the simulation. Particles are initialized with one on the left ($x_1 \approx -10$) and one on the right ($x_2 \approx 10$), producing a concentration in that region.

For joint positions to cross the diagonal, particles must pass through each other to exchange sides. While possible, such crossings are suppressed by the exclusion dynamics.

In contrast, theoretical fermionic heatmaps assume particle indistinguishability: it is fundamentally impossible to identify which particle occupies which side. Impossible to know whether the particle located at $x \approx -10$ is indeed particle 1, launched to the left, or the other particle that has switched sides. Since the particles are indistinguishable, we must assume that they may have swapped places, leading to the symmetric distribution with two lobes on both side of the diagonal.

It can easily be shown that this comes from the indistinguishability of the fermions, by varying initial conditions in my simulation, we can have either way: the spot in the top left (simulation in normal mode), the spot in the bottom right (simulation in inverted mode) or the 2 symmetrical spots as in the classical model (simulation in random mode, the particles are launched randomly P1 on the left and P2 on the right or vice versa)

Thus, fermionic indistinguishability seems to emerge here as a statistical property of ensembles of distinguishable particles, revealed through symmetry under inversion of initial conditions.

The finite value $g^{(2)}(0) \approx 0.35$, instead of the ideal $g^{(2)}(0) = 0$, is attributed to classical tunneling effects (particles with sufficient kinetic energy to overcome the repulsive barrier induced by the field), and to the absence of explicit antisymmetrization. Each crossing (red points on the other side of the heatmap diagonal), contributes directly to $g^{(2)}(r)$.

Coupling mode:

The coupling mode of the two particles interacting through the shared field can be changed in the simulation, switching between two different modes: “sum” or “diff”.

sum: $\psi = \psi_1 + \psi_2$

diff: $\psi = \psi_1 - \psi_2$

The “sum” mode induces the fermionic behavior of exclusion described above, whereas the “diff” mode induces a bosonic behavior of bunching.

Furthermore, mathematically, subtracting is equivalent to adding a wave phase shifted by π

$$\psi = \psi_1(\varphi) - \psi_2(\varphi) = \psi_1(\varphi) + \psi_2(\varphi + \pi)$$

This transforms the interference pattern from one creating an effective repulsive barrier, to one creating an effective attractive well.

This modeling could provide insights into the mechanisms underlying this quantum behavior and statistics

4. Perspectives: Emergent Polarization in 2D

Preliminary two-dimensional simulations were performed to test the robustness of the model and visualize the field structure.

- **Born rule in 2D:** The particle correctly samples the 2D field, with $\rho(x, y) \approx |\psi|^2$
- **2D structures:** Although ψ is a scalar field, analysis of the probability current $J = \text{Im}(\psi * \nabla \psi)$ reveals local vectorial structures (vortices), suggesting a possible connection with intrinsic angular momentum (spin), which remains to be explored.

Emergence of Born's rule 2D (N=6000, Steps=40000) p vs $|\psi|^2$ corr: 0.9992 | L¹: 0.02091

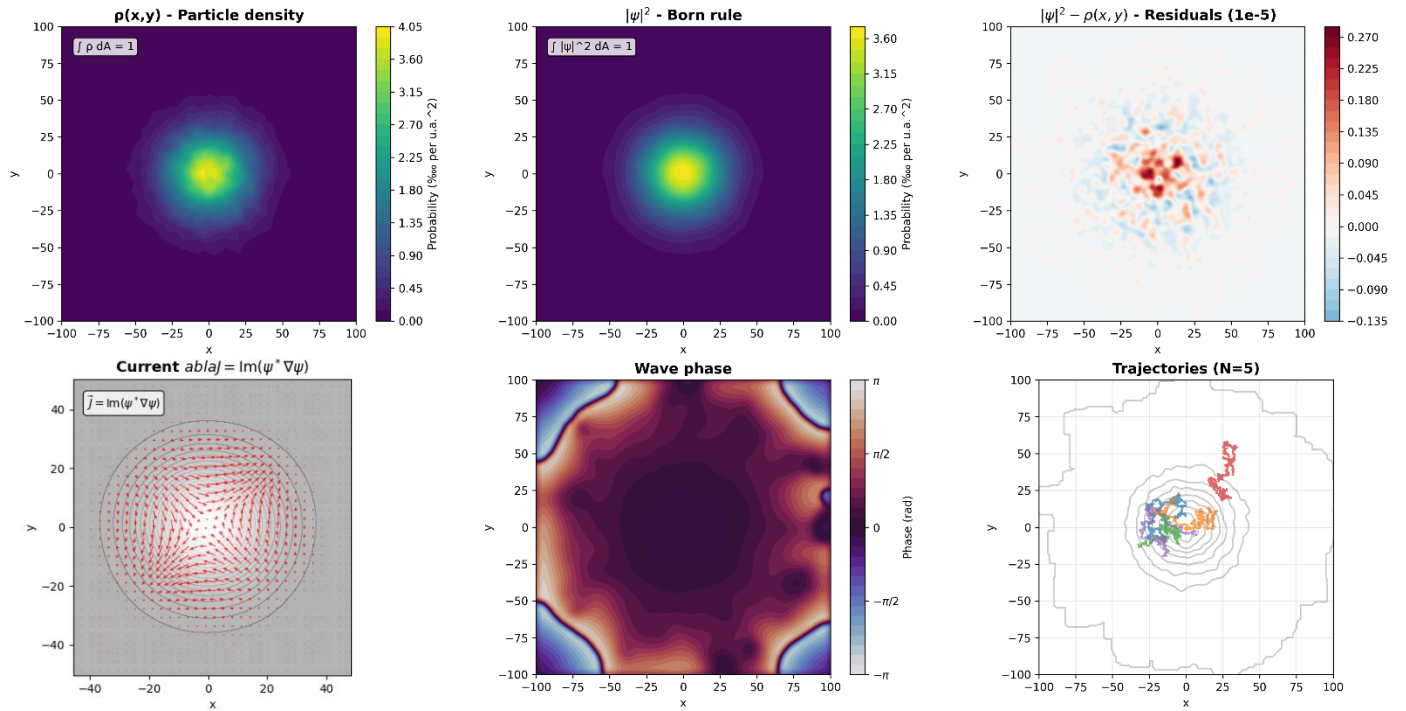


Fig 3. Simulation 2D. (Top) Comparison between particle density and Born's rule. (Bottom-left) Probability current vectors showing the internal field dynamics.

Conclusion

These results suggest that Born and Pauli statistics don't necessarily have to be taken as fundamental postulates, but may instead emerge from a local and stochastic pilot-wave dynamics with feedback. The source code (Python/Numba) is publicly available for replication.

Speculative Interpretation of the Model:

Importantly, one possible (though not mandatory) interpretation of this model is that it naturally allows for a fundamental distinction between fermions and bosons.

Fermions may be viewed as coupled particle-wave entities that interact together, whereas bosons correspond to pure waves that propagate and superpose classically. Fermions emit a bosonic excitation during each interaction with their wave field, or even continuously.

In this model, the polarization of emitted bosons (e.g., photons) can be considered as an emergent property of the propagation modes of the field emitted by the source particle. Rather than as a fundamental degree of freedom. By analogy, these modes can be likened to transverse or rotational propagation structures, evidently dependent on the orientation relative to the detector (somehow also analogous to classical wave propagation: P-waves, S-waves, etc.)